Poisson's Equation Formula Sheet

Diego Juarez **Computational Electrodynamics** **March 25, 2026**

1) Governing Equations

Electrostatics

$$\mathbf{E} = -\nabla\Phi, \qquad \nabla\cdot\mathbf{E} = \frac{\rho}{\varepsilon_0}$$

$$\boxed{\nabla^2\Phi = -\frac{\rho}{\varepsilon_0}}$$

Charge-free region:

$$\rho = 0 \Longrightarrow \nabla^2\Phi = 0$$

Linearity:

$$\Phi = a\Phi_1 + b\Phi_2 \Longrightarrow \rho = a\rho_1 + b\rho_2$$

Cartesian Laplacian:

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Spherical symmetry:

$$\nabla^2\Phi = \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\Phi}{dr}\right)$$

1D form:

$$\frac{d^2\Phi}{dx^2} = -\frac{\rho(x)}{\varepsilon_0}$$

2) Integral and Symmetry Results

Free-space solution:

$$\boxed{\Phi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\,d^3r'}$$

Point charge:

$$\rho(\mathbf{r}) = q\delta^{(3)}(\mathbf{r}), \qquad \Phi = \frac{q}{4\pi\varepsilon_0 r}$$

Radial field from Gauss's law:

$$E_r(r)\,4\pi r^2 = \frac{Q_{\mathrm{enc}}(r)}{\varepsilon_0}$$

Potential from field:

$$\Phi(r_2) - \Phi(r_1) = -\int_{r_1}^{r_2} E_r(r)\,dr$$

Uniform sphere ($r < R$):

$$E_r = \frac{\rho_0 r}{3\varepsilon_0}$$

Uniform sphere ($r > R$):

$$E_r = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}, \qquad Q = \frac{4\pi R^3}{3}\rho_0$$

3) Boundary and Uniqueness Facts

Dirichlet data: specify Φ on the boundary.
Neumann data: specify $\partial\Phi/\partial n$ on the boundary.
At an interface with no surface charge:

$$\Phi_1 = \Phi_2, \qquad \left.\frac{\partial\Phi_1}{\partial n}\right|_S = \left.\frac{\partial\Phi_2}{\partial n}\right|_S$$

With surface charge density σ:

$$E_{2n} - E_{1n} = \frac{\sigma}{\varepsilon_0}$$

Uniqueness idea:

$$\nabla^2(\Phi_1 - \Phi_2) = 0$$

and zero boundary data imply $\Phi_1 = \Phi_2$.
Physical reminder:
Potential in a region can be nontrivial even when $\rho = 0$ there, because boundary values encode external sources.

4) Useful Solvers and Checks

Finite difference

$$\frac{\Phi_{i+1} - 2\Phi_i + \Phi_{i-1}}{h^2} \approx -\frac{\rho_i}{\varepsilon_0}$$

Second-order truncation error:

$$\mathcal{O}(h^2)$$

1D jump from delta source

$$\Phi''(x) = -\frac{q}{\varepsilon_0}\delta(x-a)$$

$$\Phi(a^+) = \Phi(a^-), \qquad \Phi'(a^+) - \Phi'(a^-) = -\frac{q}{\varepsilon_0}$$

Sign check

Positive ρ gives negative curvature of Φ in 1D.

Units

$$[\Phi] = \mathrm{V}, \quad [\rho] = \mathrm{C\,m^{-3}}, \quad [\varepsilon_0] = \mathrm{C\,V^{-1}m^{-1}}$$

$$[\nabla^2\Phi] = \mathrm{V\,m^{-2}} = [\rho/\varepsilon_0]$$